Positive-actuated brake - design brief

AI BRAKING / AB-POS / C0 / 2026-09-07

A direct-applied cartridge variant of the disc concept for controlled process braking. Loss of actuation releases braking force; it does not inherit spring-applied holding behaviour.

TARGETS (not validated)

Rotor interface: 400 × 20 mm. Shared AB-ROT geometry

Study torque: 500 N·m. Process-brake hypothesis

Normal force per face: 5.38 kN. At μ=0.30, r=155 mm

Loss of pressure: Brake force lost. Must be acceptable to the machine safety concept

DESIGN DECISIONS

Reuse the rotor and selected metalwork only after confirming actuator loads.

Keep the operating principle explicit to prevent selection as a default holding brake.

Buy direct-actuation cylinders and define modulation requirements separately.

CANDIDATE MATERIALS

Shared concept bridge/pad materials with AB-DC.

Direct actuator: supplier-selected seals, body and piston materials.

Assembly and engineering gates

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ASSEMBLY INTENT

1. Confirm application permits loss of braking on loss of supply.

2. Assemble shared caliper metalwork and supplier direct actuators.

3. Set pressure limiting and independent protection after system review.

OPEN DESIGN RISKS

1. Not a fail-safe holding brake.

2. No modulation loop, thermal-duty capability or emergency-stop approval.

VALIDATION REQUIRED

1. Pressure-force/torque curve, hysteresis and response.

2. Thermal duty and loss-of-supply behaviour.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

1. Advisory position / wear sensing with replaceable brackets; no AI-controlled brake release.

2. Record supplier sensor range, supply, connector, EMC, ingress and temperature limits before selection.

3. Define independent machine safety interface and diagnostic fault response with the responsible controls engineer.

Assembly reference / numbered components

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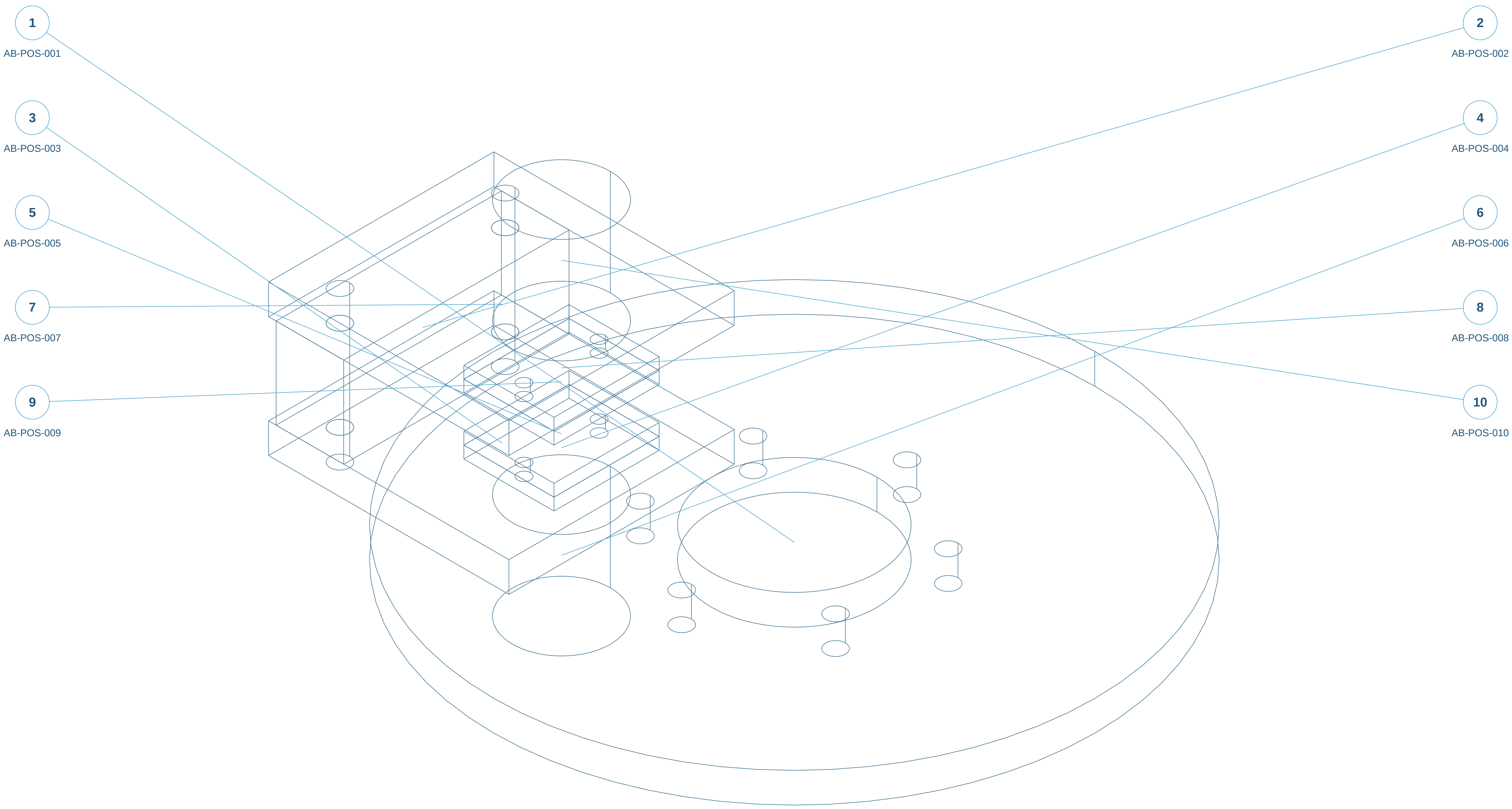

Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

AB-POS-001 | Common disc reference | shared-concept

C45 candidate; Saw/laser blank; CNC critical faces and bores

AB-POS-002 | Bridge beam | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-POS-003 | Cheek plate -1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-POS-004 | Pad backing -1 | make-concept

C45 candidate; Saw/laser blank; CNC critical faces and bores

AB-POS-005 | Friction pad -1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Purchase matched compound and approved bonding/retention

AB-POS-006 | Direct cartridge -1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Purchase cartridge; mounting interface unassigned

AB-POS-007 | Cheek plate 1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-POS-008 | Pad backing 1 | make-concept

C45 candidate; Saw/laser blank; CNC critical faces and bores

AB-POS-009 | Friction pad 1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Purchase matched compound and approved bonding/retention

AB-POS-010 | Direct cartridge 1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Purchase cartridge; mounting interface unassigned

AB-POS-001 / Common disc reference

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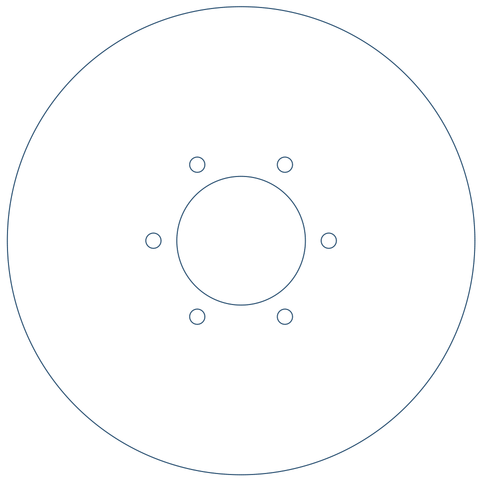
TOP X-Y



C45 candidate

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 400.00 / 400.00 / 20.00 mm

Common AB-ROT disc geometry; see rotor package for hub.

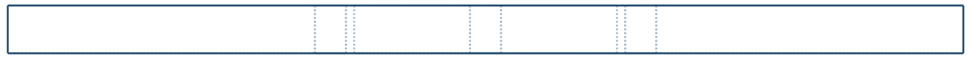
CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z

SIDE Y-Z



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AB-POS-002 / Bridge beam

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TOP X-Y

S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 150.00 / 45.00 / 60.00 mm

Bridge width 150; depth 45; axial span 60.

Tie interface bores are concept only; structural joint not released.

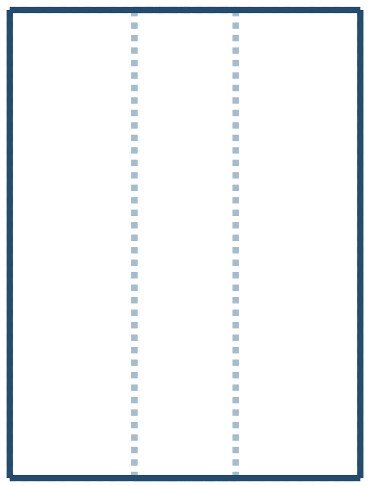
CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z

SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-POS-003 / Cheek plate -1

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S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 150.00 / 160.00 / 20.00 mm

150 x 160 x 20; mounting holes local X +/-55, Y 52.5.

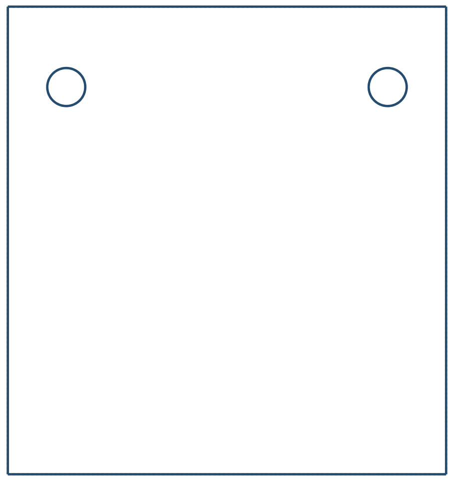
Faces mate bridge at Z +/-30; preload/fastener detail pending.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y



FRONT X-Z

SIDE Y-Z

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-POS-004 / Pad backing -1

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C45 candidate

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

70 x 60 x 8; two diameter 8.5 retainer holes.

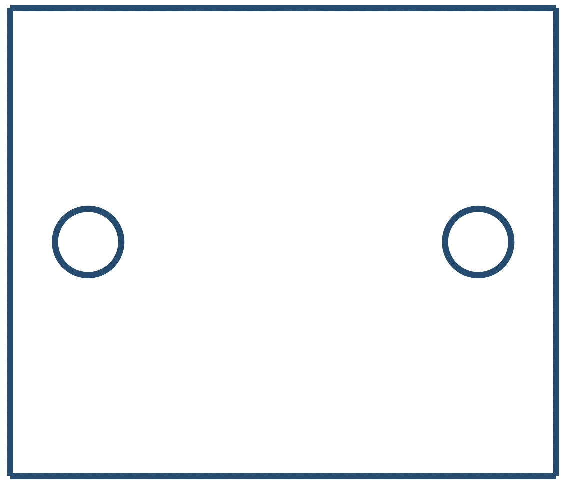
Retention detail must avoid lining damage.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

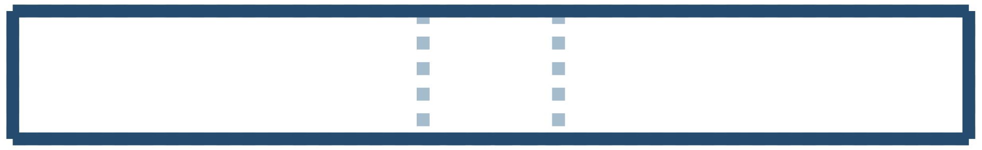
Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y



FRONT X-Z

SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-POS-005 / Friction pad -1

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Supplier-controlled; illustrative reserved envelope

Purchase matched compound and approved bonding/retention

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

Pad inner face Z +/-11; 1 mm cold study clearance to disc.

Uniform pressure effective radius approximated at 155 mm.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y

FRONT X-Z

SIDE Y-Z

AB-POS-006 / Direct cartridge -1

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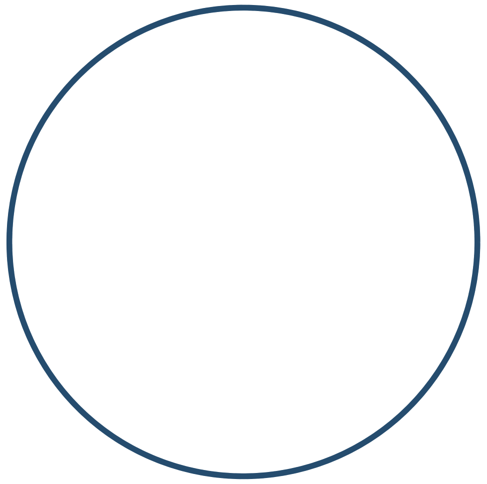
TOP X-Y



Supplier-controlled; illustrative reserved envelope

Purchase cartridge; mounting interface unassigned

Overall X/Y/Z bounds: 65.00 / 65.00 / 70.00 mm

Diameter 65 x 70 envelope only.

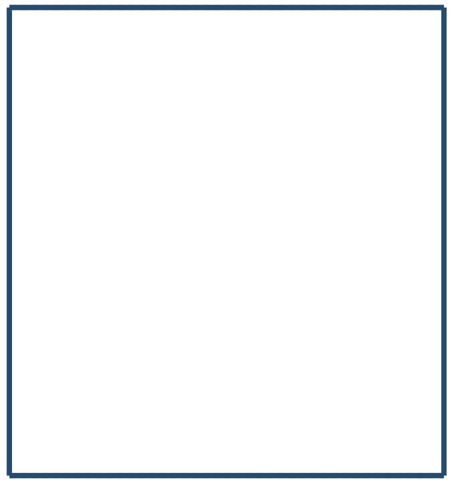
No pressure port, piston, seals or spring mechanism modelled.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

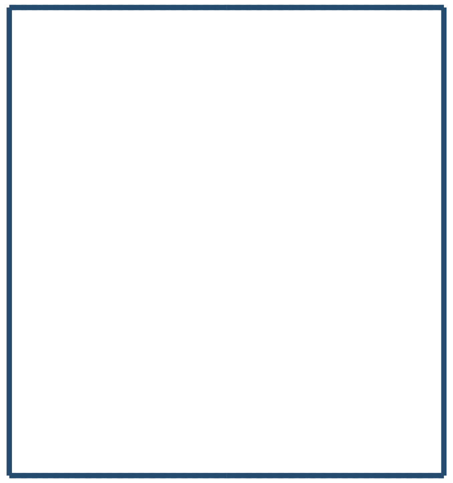
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



AB-POS-007 / Cheek plate 1

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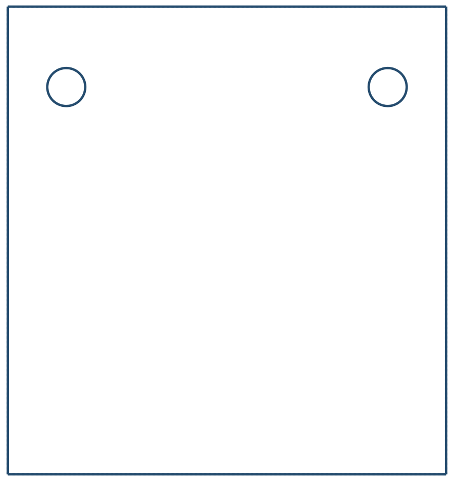
TOP X-Y



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 150.00 / 160.00 / 20.00 mm

150 x 160 x 20; mounting holes local X +/-55, Y 52.5.

Faces mate bridge at Z +/-30; preload/fastener detail pending.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z

SIDE Y-Z

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-POS-008 / Pad backing 1

AI BRAKING / AB-POS / C0 / 2026-09-07

C45 candidate

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

70 x 60 x 8; two diameter 8.5 retainer holes.

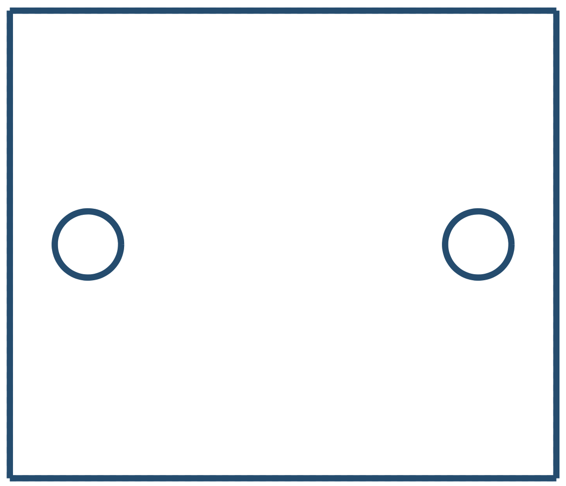
Retention detail must avoid lining damage.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

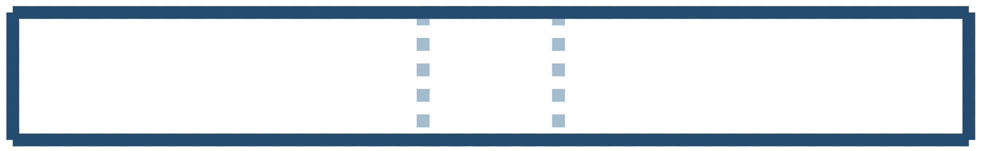
Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y



FRONT X-Z

SIDE Y-Z



AB-POS-009 / Friction pad 1

AI BRAKING / AB-POS / C0 / 2026-09-07

Supplier-controlled; illustrative reserved envelope

Purchase matched compound and approved bonding/retention

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

Pad inner face Z +/-11; 1 mm cold study clearance to disc.

Uniform pressure effective radius approximated at 155 mm.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y

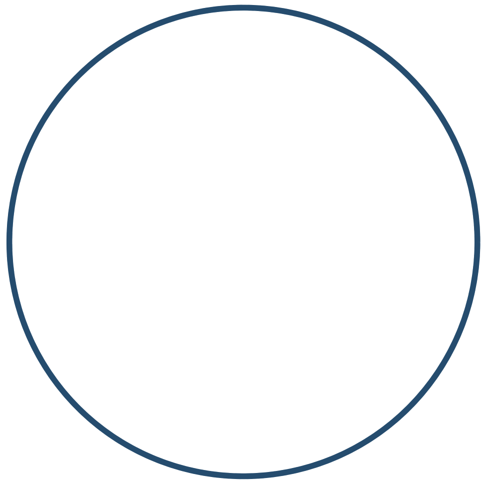
FRONT X-Z

SIDE Y-Z

AB-POS-010 / Direct cartridge 1

AI BRAKING / AB-POS / C0 / 2026-09-07

TOP X-Y



Supplier-controlled; illustrative reserved envelope

Purchase cartridge; mounting interface unassigned

Overall X/Y/Z bounds: 65.00 / 65.00 / 70.00 mm

Diameter 65 x 70 envelope only.

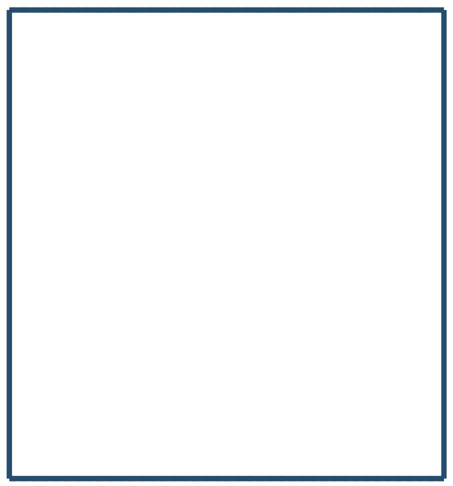
No pressure port, piston, seals or spring mechanism modelled.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z